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Optimization of Idealized Quantum Dot Intermediate Band Solar Cells Considering Spatial Variation of Generation Rates

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ABSTRACT The influences of the intermediate band (IB) filling, the absorption coefficient constants, and the IB position on the efficiency of a quantum dot intermediate band solar cell (QD-IBSC) are investigated considering the spatial variation of subbandgap generation rates. A new definition of optimal intermediate band filling is proposed. A mathematical model is developed to optimize the intermediate band solar cell (IBSC) structure under idealized conditions, which calculates the optimal ratio of the subbandgap absorption coefficient constants and the optimal position of IB.

INDEX TERMS Intermediate band solar cell (IBSC), conversion efficiency, optimal filling, absorption coefficients, IB position.

I. INTRODUCTION

Solar energy is perceived to be the only sustainable source of energy for the planet. Photovoltaics (PV) is the most attractive technology to convert solar energy into useful forms. Although the solar energy obtained via solar cells is presently more expensive than the energy from fossil fuels, the cost of solar cells per unit electric power output is decreasing at an exponential rate [1]. Combined with the increasing cost of fossil fuels, grid parity of solar energy is expected to be achieved before the end of this decade.

In order to continue the cost reduction of solar energy from PV, it is necessary to produce solar cells with higher efficiencies [2]. The conversion efficiency of a single p-n junction solar cell is bounded by the Shockley-Queisser limit [3], whose value for direct conversion is 41% (with 31% global conversion efficiency limit). For this reason, the focus of the third generation PV is to exceed the Shockley-Queisser limit of the conversion efficiency [4]. Tandem solar cells are so far the most promising technology for high efficiency solar cells offering a direct conversion efficiency limit of nearly 89% for infinite number of cascaded cells [5]. Intermediate band solar cells (IBSC) also have a high conversion efficiency limit with 63% for direct conversion [6] which is almost equal to the efficiency limit of a three cell tandem system. In addition IBSC should offer superior performance in terms of interconnectivity and spectral sensitivity [4]. Therefore, IBSC are attracting a lot of attentions as a potential third generation PV technology.

IBSC employs a band of electron states within the bandgap to create additional optical transition paths in order to provide higher conversion efficiency. The solar spectrum is used more efficiently in the IBSC as the subbandgap photons are utilized for optical transitions corresponding to the intermediate band (IB). In addition to the usual optical transition from valence band (VB) to conduction band (CB), successive transitions from VB to IB and from IB to CB (as shown in Fig. 1) can contribute to the photocurrent which boosts the efficiency of the cells.

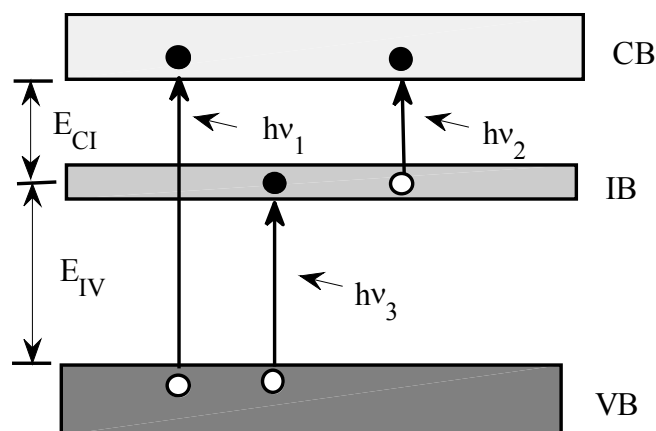


FIGURE 1. Simplified schematic of optical transitions in an intermediate band solar cell (IBSC).

Several approaches have been proposed to incorporate the basic principles of intermediate band solar cells including quantum dots (QD) [7]–[12], impurity photovoltaic effect [13], [14] and dilute semiconductor alloys [15]–[17]. InAs quantum dots in GaAs has emerged as a promising system for implementing the concept of IBSC. Quantum dot intermediate band solar cell (QD-IBSC) prototypes have been fabricated by introducing intermediate states in the band gap of GaAs through InAs QD layers [18]–[23]. In most devices, the conversion efficiencies of the IBSC were found to be less than the efficiencies of the corresponding baseline GaAs cells. Open circuit voltage degradation due to overlapping VB sub-bands of the QDs was responsible for this phenomenon. Recently, open circuit voltage degradation has been minimized by adopting innovative fabrication methods [21], [23]. In spite of that, the conversion efficiency of the IBSC was still lower than that of the baseline cell in [21] or was improved only marginally in [23] due to less than expected photocurrent enhancement.

Optimal design of the IBSC is necessary for achieving higher efficiency. Non-radiative effects also need to be suppressed. For the proper operation of the IBSC, the IB has to be partially filled [7], [8]. The occupation of the IB states is controlled by the doping density of the delta-doped GaAs layers in between successive layers of InAs QDs. The partial filling of the intermediate band ensures that the IB has sufficient number of empty states and filled states in order to accommodate electrons from the VB and to provide electrons to the CB respectively. Recently the effect of IB occupation on cell efficiency has been investigated and the theoretical optimal efficiency has been found for a number of IB devices [24]–[26]. An analytical expression for optimal filling is obtained for uncoupled IB-CB and wide IB region by Luque and Marti [27]. The reported results are obtained using the assumption of sufficiently weak subbandgap absorption coefficients so that the subbandgap optical generation rates are not spatially varying [28].

The values of the absorption coefficients can be varied by changing the QD size and shape. Zhang and Galbraith reported the value of IB-CB absorption coefficient of InAs/GaAs QDs to be about $(1-4) \times 10^4 \text{ cm}^{-1}$ [29]. Absorption coefficients that belong to this range can be considered fairly strong, i.e. the spatial variation of the generation rates cannot be neglected. Yoshida *et al.* have recently investigated the effects of doping the IB region and of changing the incident light concentration using a simulation model that considers the spatial dependence of the generation rates [30]. The previously reported analytical expression of [27] for optimal filling is not applicable for spatially varying generation rates. We have recently studied the effect of IB filling on the efficiency and have observed that the maximum efficiency is achieved around the filling that maximizes the short-circuit current [31]. The IBSC device structure is optimized in previously published literatures by finding the optimal ratio of absorption coefficient constants and the optimal position of IB via numerical calculations [25]–[28], [32].

The present study investigates the limitations of the existing definition of optimal filling [27] and proposes a new definition that incorporates the spatial variation of the generation rates. The proposed definition captures the effects of the variation of the magnitudes of the absorption coefficient constants and the base width. Moreover, an insight into the device electrostatics at optimal IB filling is provided. We also introduce an analytical optimization model which optimizes the IBSC by optimizing three device parameters, i.e. the IB filling, the ratio of the absorption coefficient constants and the IB position.

II. MODEL DESCRIPTION

The base of IBSC structure used in this study consists of multiple InAs quantum dot (QD) layers in GaAs. Fig. 2 shows a schematic diagram of the device under consideration. The QDs introduce several confined electron- and hole-states in the base region among which only the fundamental electron states serve as the IB. The base is sandwiched between a p-emitter and an n-emitter, where both the emitters are constructed of GaAs.

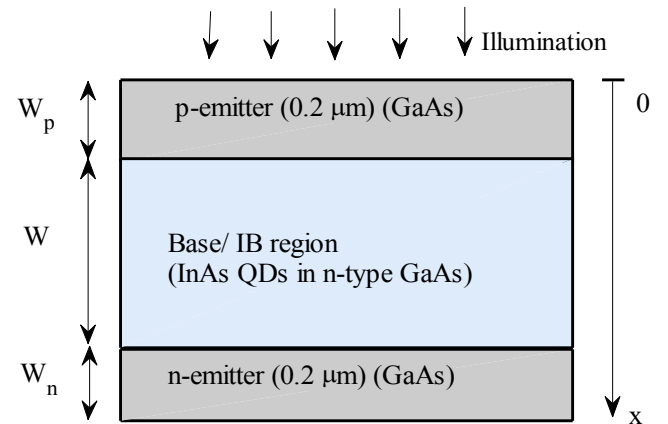


FIGURE 2. Structure of the InAs/GaAs QD-IBSC device considered in this work.

The device characteristics are simulated by self-consistently solving one dimensional carrier continuity equations, Poisson's equation and balance equation of intermediate band using the drift-diffusion model presented by Lin and Phillips [33]. Radiative recombination is considered to be the dominant recombination mechanism while non-radiative recombinations are ignored. Thermal generation is also neglected for simplicity, hence the bands are considered thermally uncoupled. The optical generation and radiative recombination terms are given by [34]

$$G_{ij} = \int_E \alpha_{ij} F_0 \exp(-\alpha_{ij} x) dE \quad (1)$$

$$R_{ji} = \frac{8\pi}{h^3 c^2} \int_E \alpha_{ij} E^2 \exp\left(-\frac{E}{kT}\right) \exp\left[\left(\frac{\mu_{ji}}{kT}\right) - 1\right] dE \quad (2)$$

where G_{ij} is the optical generation rate, R_{ji} is the radiative recombination rate and α_{ij} is the absorption coefficient. The

TABLE 1. Simulation parameters: nomenclature and values.

Symbol	Description	Value
E_g	Bandgap energy	1.42 eV
E_{CI}	CB-IB subbandgap energy	
E_{IV}	IB-VB subbandgap energy	
N_c	Density of states in CB	$4.7 \times 10^{17} \text{ cm}^{-3}$
N_v	Density of states in VB	$7 \times 10^{18} \text{ cm}^{-3}$
N_t	Density of states in IB	$4 \times 10^{17} \text{ cm}^{-3}$
N_A	p-emitter doping	$4 \times 10^{17} \text{ cm}^{-3}$
N_D	n-emitter doping	$4 \times 10^{17} \text{ cm}^{-3}$
W_n, W_p	n-emitter and p-emitter width	0.2 μm
μ_n	Electron mobility [35]	$3000 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$
μ_p	Hole mobility [35]	$700 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$
T	Cell temperature	300 K
T_{sun}	Sun temperature	5963 K
ϵ_r	Relative permittivity	13
X	Solar concentration	1000 suns
α_{VC}	VB to CB absorption coefficient	$2 \times 10^4 \text{ cm}^{-1}$

subscript $ij = IC, VI, VC$ correspond to IB-CB, VB-IB, VB-CB transitions respectively. μ_{ji} is the corresponding quasi-Fermi level split between the bands. F_0 is the incident photon flux derived from the Planck formula providing the spectrum of blackbody radiation at 5963 K.

When the density of states in IB is finite and the transport through the intermediate band is neglected, the generation-recombination terms associated with the IB has to satisfy the IB balance equation represented by Eq. (3) at each value of x .

$$G_{IC}(x) - R_{CI}(x) = G_{VI}(x) - R_{IV}(x). \quad (3)$$

The contacts are assumed to be ideal ohmic contacts with infinite surface recombination velocity. The absorption coefficients, α_{ij} governing the subbandgap optical generation and radiative recombination terms are dependent on the occupation of the intermediate band states, f . The absorption coefficients corresponding to IB-CB and VB-IB transitions are given by

$$\alpha_{IC} = \alpha_{IC0}f, \quad \alpha_{VI} = \alpha_{VI0}(1 - f) \quad (4)$$

where, α_{IC0} is the value of IB-CB absorption coefficient corresponding to optical transitions between a completely filled IB and CB while α_{VI0} relates to optical transitions between an entirely empty IB and VB. The three absorption coefficients (α_{VC} , α_{IC} and α_{VI}) are considered spectrally non-overlapping. The parameters used in this work are listed in Table I.

III. OPTIMIZATION OF QD-IBSC

The conversion efficiency of a solar cell, η is dependent on the short-circuit current density J_{SC} , open-circuit voltage, V_{OC} and fill-factor, FF of the cell through the relation

$$\eta = \frac{J_{SC} \times V_{OC} \times FF}{P_{\text{sun}}}. \quad (5)$$

The efficiency of the IBSC can be controlled not only by the IB filling but also by the values of absorption coefficient

constants and the position of the IB within the bandgap. Strandberg and Reenas have reported the optimum value of $\alpha_{IC0}/\alpha_{VI0}$ ratio while keeping the geometrical average constant [25]. Hu *et al.* have optimized the IB position and $\alpha_{IC0}/\alpha_{VI0}$ ratio by varying these independently in numerical simulations and have reported the requirement of a large value of α_{IC0} for maximizing efficiency [32]. In this study we present a mathematical model for finding the optimized values of IB filling, $\alpha_{IC0}/\alpha_{VI0}$ ratio and IB position for QD-IBSC when the spatial variation of the subbandgap generation rates is not negligible.

A. OPTIMAL FILLING

It is observed in our previous study that the short-circuit current density undergoes the largest changes when the IB occupation is varied via base doping, while the effects on the fill factor and the open-circuit voltage are comparatively small [31]. Hence, the efficiency follows the trend of the short-circuit current variation when the IB occupation is changed as shown in Fig. 3. Consequently, the maximum efficiency occurs approximately at the IB filling that provides the maximum short circuit current. In order to achieve the maximum efficiency, the short-circuit current is required to be maximized which can be achieved by maximizing the net generation rate ($G_{ij} - R_{ji}$). This hypothesis has also been used in [30]. The increase of the IB occupation boosts G_{IC} while reducing G_{VI} , while the reducing IB occupation has opposite effect. As the net generation is limited by the smaller subbandgap generation rate, the most efficient transition of electrons from the VB to the CB can be ensured when the subbandgap generation rates match. Thus the condition for obtaining the maximum generation is as in Eq. (6).

$$G_{IC} = G_{VI}. \quad (6)$$

At this condition, the recombination terms should be matched too in order to satisfy the IB balance condition in Eq. (3).

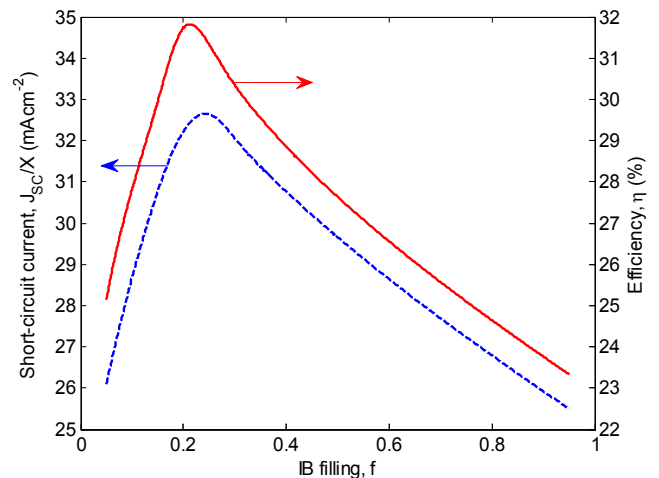


FIGURE 3. Variation of short-circuit current, J_{SC} and conversion efficiency, η with IB filling, f . $\alpha_{IC0} = \alpha_{VI0} = 1 \times 10^4 \text{ cm}^{-1}$, $E_{CI} = 0.3 \text{ eV}$, $W = 0.5 \mu\text{m}$.

The optical generation processes can be expressed by [27]

$$G_{IC} = \gamma_{IC} N_t f v_{th} \quad (7)$$

$$G_{VI} = \gamma_{VI} N_t (1 - f) v_{th} \quad (8)$$

where γ_{IC} and γ_{VI} are parameters proportional to the light intensity accounting for the photons absorbed by optical transitions and v_{th} is the thermal velocity of the electrons. γ_{IC} and γ_{VI} are related to the absorption coefficients by

$$\gamma_{IC} = \frac{1}{N_t v_{th}} \int_E \alpha_{IC0}(E) F_0(E, x) dE \quad (9)$$

$$\gamma_{VI} = \frac{1}{N_t v_{th}} \int_E \alpha_{VI0}(E) F_0(E, x) dE. \quad (10)$$

In previous works γ_{IC} and γ_{VI} were considered to be constants which makes the generation terms G_{IC} and G_{VI} independent of x [27], [28]. As a result, G_{IC} and G_{VI} are perfectly matched at all values of x at the maximum generation condition. The IB filling that satisfies this condition is identified as the optimal filling by Luque *et al.* [27] which is expressed as

$$f_{opt} = \frac{\gamma_{VI}}{\gamma_{VI} + \gamma_{IC}}. \quad (11)$$

However, γ_{IC} and γ_{VI} are independent of x only if the dependence of $F_0(E, x)$ on x is negligible, i.e. if $\alpha_{ij} \ll 1$. Moreover, $F_0(E, x)$ and consequently γ_{IC} and γ_{VI} are functions of absorption coefficient, α_{ij} i.e. functions of f . As a result, Eq. (11) is not applicable for spatially varying generation terms. When we numerically solve Eq. (6) for f , the solution is found to depend on x , as shown in Fig. 4. The variation is more pronounced for larger α_{ij0} . The IB filling that matches subbandgap generations at the front end of the base can produce large mismatch at the back end. Hence it is necessary to find the exact conditions required for optimal efficiency in case of spatially varying generation rates.

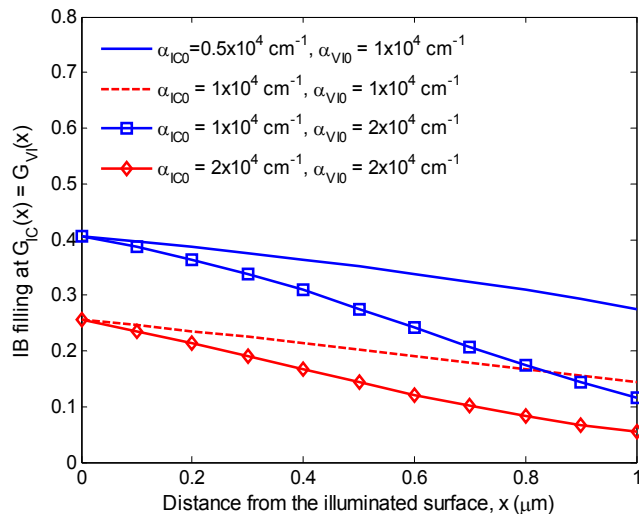


FIGURE 4. IB filling required for matching the $G_{IC}(x)$ and $G_{VI}(x)$ with $E_{CI} = 0.3$ eV.

In case of spatially varying generation rates, optimal filling, f_{opt} may be determined from Eq. (12).

$$\int_{W_p}^{W+W_p} G_{IC}(x, f_{opt}) dx = \int_{W_p}^{W+W_p} G_{VI}(x, f_{opt}) dx. \quad (12)$$

When $\alpha_{ij} \ll 1$, the spatial variation of generation rates becomes negligible and f_{opt} in Eq. (12) approaches the value given by Eq. (11). For step-like absorption coefficients, we express γ_{IC} and γ_{VI} as

$$\gamma_{IC} = \alpha_{IC0} \phi_L / N_t v_{th}, \quad \gamma_{VI} = \alpha_{VI0} \phi_H / N_t v_{th} \quad (13)$$

where

$$\phi_H = \int_{E_{IV}}^{E_g} F_0(E) dE, \quad \phi_L = \int_{E_{CI}}^{E_{IV}} F_0(E) dE. \quad (14)$$

Strong spatial variation of subbandgap generation can be due to either large values of α_{ij0} or large W . As values α_{ij0} and W are varied, the validity of Eq. (12) is checked next.

B. VARIATION OF ABSORPTION COEFFICIENT CONSTANTS

The peak short-circuit current is larger for stronger absorption coefficient constants, α_{IC0} and α_{VI0} , as the subbandgap generation rates are increased and the photocurrent is boosted. Although the IB filling does not have a significant effect on the open circuit voltage, V_{OC} [31], larger values of α_{IC0} and α_{VI0} improve V_{OC} slightly. The improvement is the result of a larger amount of subbandgap optical generation, which widens the energy split between electron quasi-Fermi level and hole quasi-Fermi level. The fill factor, FF is generally found to increase for stronger absorption coefficients. As a result, larger values of α_{IC0} and α_{VI0} enhance the conversion efficiency, which is evident from Fig. 5. The optimal filling moves towards higher values for a lower $\alpha_{IC0}/\alpha_{VI0}$ ratio, which can be explained by the fact that G_{IC} and G_{VI} are functions of $\alpha_{IC0}f$ and $\alpha_{VI0}(1-f)$ respectively. It is also observed

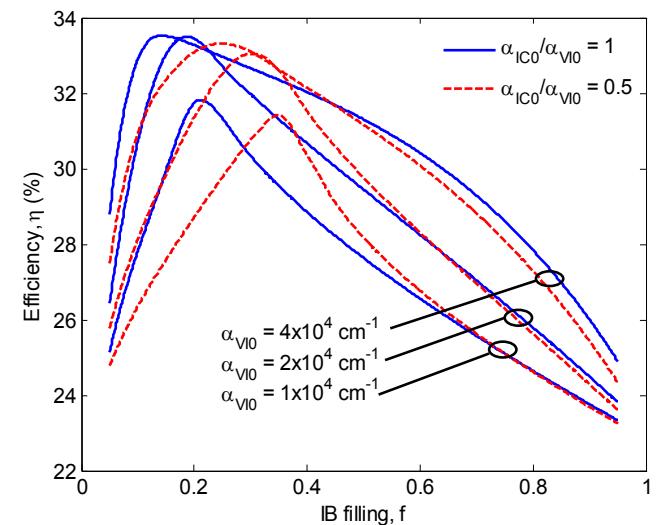


FIGURE 5. Variation of conversion efficiency, η with IB filling, f for different values of α_{IC0} and α_{VI0} , $E_{CI} = 0.3$ eV, $W = 0.5$ μm .

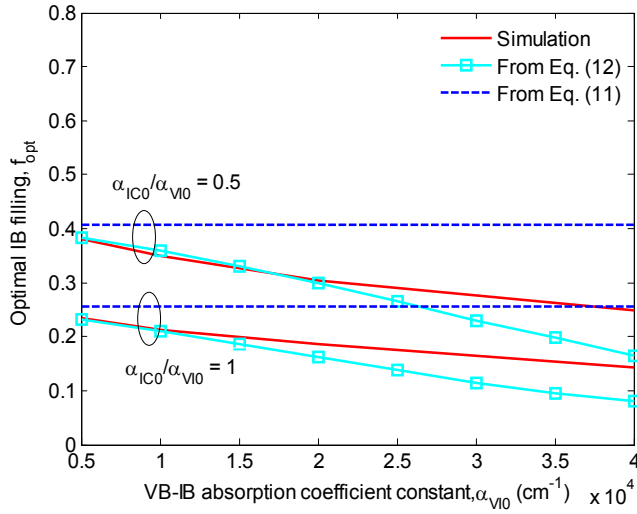


FIGURE 6. Optimal IB filling as a function of absorption coefficient constants. $E_{Cl} = 0.3$ eV, $W = 0.5$ μm .

that around the peak value, the efficiency becomes less sensitive to the changes in IB filling when α_{IC0} and α_{VI0} are larger.

Fig. 6 shows the variation of optimal filling, f_{opt} with absorption coefficient constants. The numerical simulation results are compared to both the solutions obtained from Eq. (11) and Eq. (12). Simulation results show that the optimal IB filling decreases with the increase of the absorption coefficient constants, also shown by Eq. (12). The solution from Eq. (11) is unable to capture this trend and consistently over-estimates f_{opt} . The results obtained from Eq. (12) deviate from the simulation results at very high values of absorption coefficient constants. Nevertheless, since a small shift from the optimal filling has a less significant effect on the performance of a cell with stronger absorption coefficients (Fig. 5), the efficiency of the cell will not suffer much if the IB filling is set to the value predicted by Eq. (12) even for large values of α_{ij0} .

C. VARIATION OF BASE WIDTH

Fig. 7 shows the relationship between the conversion efficiency and IB filling for different base widths. The effects of increasing the base width are similar to the effects of increasing α_{IC0} and α_{VI0} . A wider base increases the total amount of generation in the base, leading to larger J_{SC} and V_{OC} . FF also increases for wider bases. Hence the efficiency is boosted when base width increases.

The optimal filling found from the simulation results is compared to both the definitions of optimal filling in Fig. 8. It can be observed from Fig. 6 and Fig. 8 that the estimation of f_{opt} from Eq. (12) is accurate as long as $\alpha_{ij0}W \leq 1$.

D. BAND DIAGRAMS AT OPTIMAL FILLING

The base region of an IBSC can behave as a n-type region or a p-type region depending on the IB filling [27]. If $f > f_{opt}$, the base resembles an n-type region, whereas it behaves as a

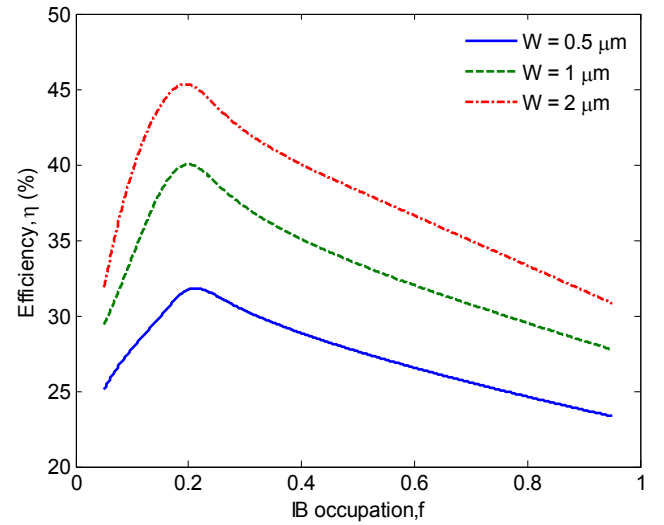


FIGURE 7. Variation of conversion efficiency, η with IB filling, f for different base widths, W . $E_{Cl} = 0.3$ eV, $\alpha_{IC0} = \alpha_{VI0} = 1 \times 10^4$ cm^{-1} .

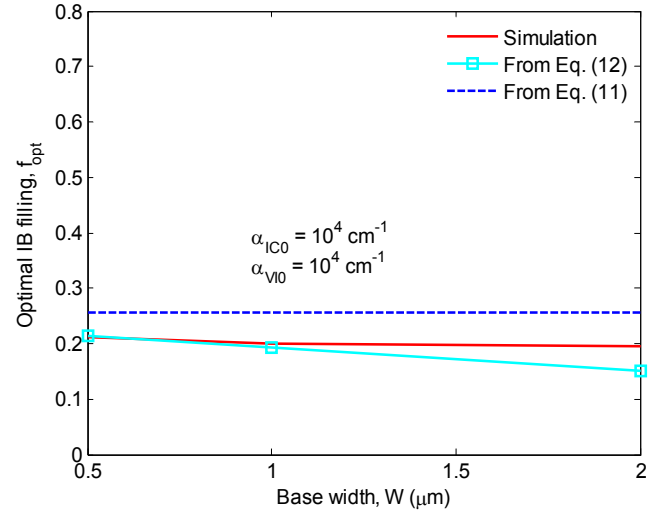


FIGURE 8. Variation of optimal IB filling with base width, W . $E_{Cl} = 0.3$ eV.

p-type region for $f < f_{opt}$. When $f = f_{opt}$, the band-diagram of the base will depend on the position of IB. It is to be noted that, G_{IC} is not equal to G_{VI} at all points inside the base region at optimum IB occupation unless ϕ_L is equal to ϕ_H . If the position of IB makes ϕ_L equal to ϕ_H , the base region behaves as an intrinsic region at optimal filling, i.e. the whole structure behaves as a p-i-n structure. Fig. 9(a) illustrates the calculated short-circuit condition band diagram for $f = f_{opt}$ when $\phi_L = \phi_H$, i.e. both the generation rates match at all points in the base region. In this case, both Eq. (11) and Eq. (12) can predict the correct optimal filling for any values of α_{ij0} and W . On the other hand, G_{IC} and G_{VI} are equal only at a single point inside the base region at maximum efficiency condition if $\phi_L \neq \phi_H$ (Fig. 9(b)). At this condition, G_{IC} is less than G_{VI} in the part of the base adjacent to the p-emitter, so this region behaves as

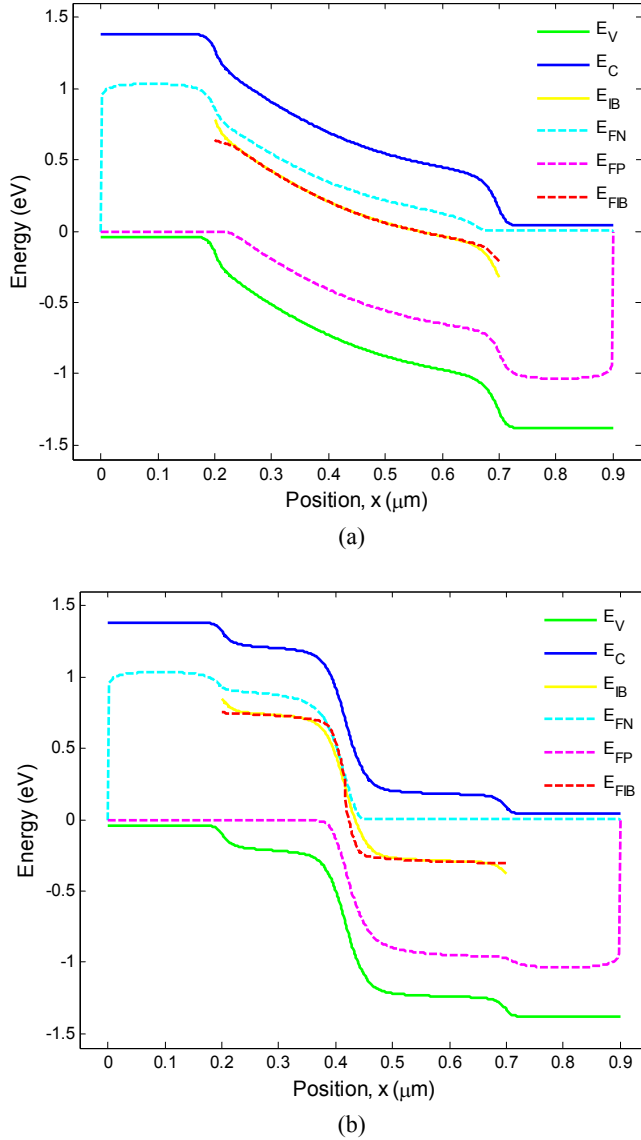


FIGURE 9. Band diagram for QD-IBSC with optimal filling when generation rates, G_{IC} and G_{VI} (a) match everywhere within the base, $E_{CI} = 0.484$ eV and (b) do not match everywhere within the base, $E_{CI} = 0.47$ eV. $\alpha_{IC0} = \alpha_{VI0} = 1 \times 10^4 \text{ cm}^{-1}$.

a p-region. Meanwhile, the rest of the base behaves as an n-region since G_{IC} is greater than G_{VI} at this region. As a result, the whole IBSC structure acts as a p-p⁻-n⁻-n structure. As the difference between ϕ_L and ϕ_H increases, the optimal filling found from Eq. (11) deviates farther from the simulation results.

E. OPTIMAL RATIO OF ABSORPTION COEFFICIENT CONSTANTS

After the optimal filling is determined, $\alpha_{IC0}/\alpha_{VI0}$ ratio is optimized keeping the IB filling at the optimum level, f_{opt} . The geometric mean of α_{IC0} and α_{VI0} is taken as $1 \times 10^4 \text{ cm}^{-1}$ while the ratio is varied.

The absorption coefficient constants can be expressed as

$$\alpha_{IC0} = \alpha\sqrt{r}, \quad \alpha_{VI0} = \alpha/\sqrt{r} \quad (15)$$

where α is the geometric mean of α_{IC0} and α_{VI0} and r is their ratio. Fig. 10 shows the efficiency contour plots of a QD-IBSC with a base width (W) of $0.5 \mu\text{m}$ and $E_{CI} = 0.47$ eV. It is evident that the optimal filling shifts towards smaller values as the $\alpha_{IC0}/\alpha_{VI0}$ ratio, r becomes larger. The efficiency achieved at optimal IB filling is dependent on the value of r for a certain E_{CI} . The efficiency at optimal filling is found to reach a peak value of 33.1% for $r = 0.9$. Hence the optimal ratio of absorption coefficient constants, r_{opt} is 0.9 for $E_{CI} = 0.47$ eV. Similar efficiency contour plots are calculated and values of r_{opt} are determined for different values of E_{CI} between 0 and $E_g/2$. Fig. 11 shows the variation of efficiency at optimal filling, $\eta(f_{opt}, r, E_{CI})$ with the $\alpha_{IC0}/\alpha_{VI0}$ ratio, r for different values of E_{CI} .

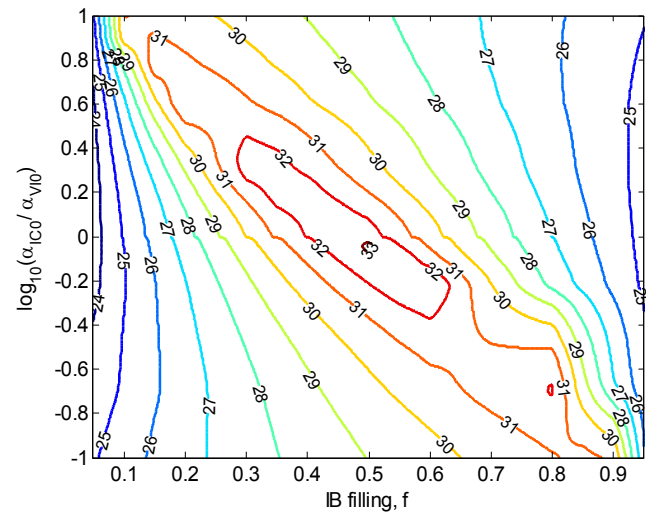


FIGURE 10. Efficiency contour plot for QD-IBSC with $E_{CI} = 0.47$ eV.

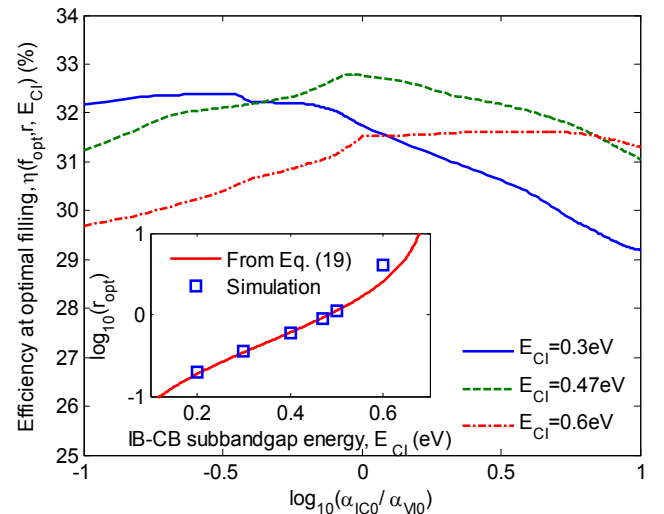


FIGURE 11. Variation of efficiency at optimal filling with the $\alpha_{IC0}/\alpha_{VI0}$ ratio for different positions of IB with $W = 0.5 \mu\text{m}$. Inset: Comparison between the proposed model (Eq. (19)) and simulation in terms of optimal ratio, r_{opt} .

When the IB filling is adjusted to the optimal value, f_{opt} the conversion efficiency, $\eta(f_{opt}, r, E_{CI})$ is a function of r

and E_{CI} . The optimal values of r and E_{CI} are required to maximize the net optical generation for $f = f_{opt}$. At this filling, the subbandgap generation rates, G_{IC} and G_{VI} are approximately equal at the p-emitter end of the base, i.e.

$$G_{IC}(W_p, f_{opt}) = G_{VI}(W_p, f_{opt}). \quad (16)$$

In order to maximize the net generation, we need to maximize either of the subbandgap generation rates in Eq. (16) by choosing the appropriate values of r and E_{CI} . With the help of Eqs. (1), (11), (13) and (14), $G_{IC}(W_p, f_{opt})$ can be expressed as

$$G_{IC}(W_p, f_{opt}) = \frac{\alpha_{IC0}\alpha_{VI0}\phi_L\phi_H}{\alpha_{IC0}\phi_L + \alpha_{VI0}\phi_H} \exp(-\alpha_{IC0}f_{opt}W_p). \quad (17)$$

Since W_p is small, $\exp(-\alpha_{IC0}f_{opt}W_p) \approx 1$. Using Eq. (15), Eq. (17) can be simplified to

$$G_{IC}(W_p, f_{opt}, r) = \frac{\alpha\phi_L\phi_H\sqrt{r}}{r\phi_L + \phi_H}. \quad (18)$$

The value of r that maximizes $G_{IC}(W_p, f_{opt})$ is given by

$$r_{opt} = \phi_H / \phi_L. \quad (19)$$

r_{opt} is found to be 0.91 for $E_{CI} = 0.47$ eV from Eq. (19), which is in good agreement with the optimal ratio obtained from simulation. The inset in Fig. 11 shows that the analytical values of r_{opt} from Eq. (19) agree well with the simulation results for different values of E_{CI} .

F. OPTIMAL POSITION OF IB

The peak efficiency obtained at optimal $\alpha_{IC0}/\alpha_{VI0}$ ratio and optimal IB filling also depends on the value of E_{CI} . If the value of E_{CI} increases, i.e. the IB moves towards the valence band, it boosts ϕ_H and diminishes ϕ_L . So the position of the IB needs to be optimized in order to determine the maximum efficiency that can be obtained from a given structure. The simulation results in Fig. 12 shows the variation of $\eta(f_{opt}, r_{opt}, E_{CI})$ with E_{CI} which has a peak occurring at $E_{CI} = 0.484$ eV, which can be regarded as the optimal position of the IB. It is notable that $E_{CI} = 0.484$ eV corresponds to perfect spatial matching of generation rates for the InAs/GaAs QD-IBSC (Fig. 9(a)) under our consideration.

The optimal value of E_{CI} corresponding to the best matched generation rates can be found by solving

$$\phi_L = \phi_H \quad (20)$$

for E_{CI} . The theoretical value of optimal E_{CI} found from Eq. (20) is 0.484 eV. According to the simulation results, maximum efficiency of 33.2% is obtained for $E_{CI, opt} = 0.484$ eV at ideal values of $r_{opt} = 1$ and $f_{opt} = 0.5$.

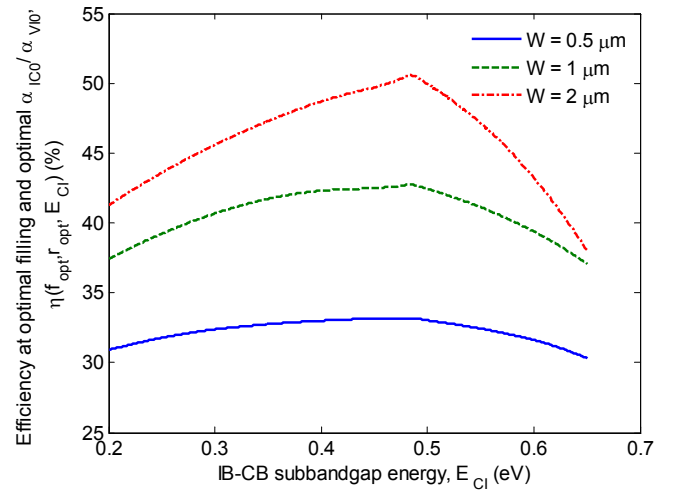


FIGURE 12. Variation of efficiency at optimal filling and optimal ratio with the positions of IB.

IV. CONCLUSION

The optimal IB filling becomes a function of the magnitudes of the absorption coefficient constants and the width of the IB region when the spatial variation of the subbandgap generation rates is not negligible. A new definition of the optimal filling is proposed taking into account this spatial variation. Through numerical simulation, it is shown that the proposed definition is valid when $\alpha_{ij0}W \leq 1$. Moreover, we find that the theoretical condition of the matched subbandgap generation rates is not actually satisfied at all points in the base region for optimally filled IB unless the subbandgap photon fluxes satisfy the condition $\phi_L = \phi_H$. The base behaves as an intrinsic region at optimal filling for $\phi_L = \phi_H$. Otherwise, at optimal filling the base acts as a p⁻n⁻ region in a p-p⁻n⁻n structure. A mathematical model is proposed to optimize the QD-IBSC structure by finding the optimal values of the IB filling, the ratio of the absorption coefficient constants and the position of the IB keeping the geometric mean of the absorption coefficient constants fixed. When the condition of $\phi_L = \phi_H$ is achieved through selection of the optimal value of E_{CI} , a structure with ideal values of $f = 0.5$ and $r = 1$ will yield the maximum efficiency.

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